

Develop AI Applications with Foundation Models



Estimated time needed: 45 minutes

Developing AI applications with generative AI foundation models

Given excellent natural language understanding and generation, the foundation models can serve as a base for you to develop applications for tasks like text completion, translation, summarization, etc. Their power and distinguishing features can directly impact the feasibility, effectiveness, and efficiency of your AI application.

Developing AI applications with generative models is an ongoing process that may require iteration and adaptation based on user needs and changing requirements. It involves several steps, beginning with understanding the use case and clearly defining the problem you aim to solve with your AI application.

For this, you must determine the generative AI foundation model that best suits your use case. This will help you make informed decisions about the direction of your AI application development.

In this lab, you will leverage foundation models' capabilities that exhibit comprehensive and coherent content creation ability in the context of solutions to a specific problem.

Learning Objectives:

1. Explore the vocabulary generation using GPT-5 Nano to develop an AI app like a dictionary
2. Explore language learning capabilities using GPT-5 Nano to develop an educational tool or a language learning app
3. Experiment with the translation capabilities of the foundation model GPT-5 Nano to create an AI-powered app for translation services

About generative AI classroom lab

► [Click here](#)

Notes:

1. The prompts used in this lab are for your reference only. You can create your own prompts and generate responses using generative AI.
2. Since AI-generated outputs are dynamic, you may receive different responses even though you've used the same prompt from this lab.

How can you develop AI applications with generative AI foundation models?

Exploring various foundation models serves as a starting point for various AI applications. Understanding and examining their capabilities will help you draw a roadmap for developing AI applications.

Using an appropriate foundation model ensures you choose the right tools and technologies, saving time, effort, and resources.

In this lab, you will explore various foundation models by finding answers to specific questions, such as:

- How can you develop an AI app that creates a scenario or object-specific content?

You can leverage the text generation capability of generative AI foundation models to create moral stories, academic poems, social media content, product descriptions, and more that can extend the scope to fields like sentiment analysis.

Before you begin prompting

Before beginning the exercises, you must set up the AI classroom and know the Generative AI workspace.

1. As a first step, you must set up your AI classroom for better learning experience.

- 1. Name the chat
- 2. Choose the foundation model

2. Know your workspace to prompt the foundation model-based chatbot better.

- 1. Use prompt instructions
- 2. Type your message
- 3. Manage your chat
- 4. Delete the chat from history

How GPT-5 Nano can help develop AI applications

OpenAI's GPT-5 Nano can help you generate text in different languages, assist in language translation, grasp language with detailed guides and explanations, and respond to your language-related queries.

Combining AI power with traditional learning methods, the model excels in language understanding tasks and can certainly aid you in developing AI applications around them.

It can be commercially used for various applications focused on NLP tasks, like:

- Language understanding and generation
- Vocabulary expansion
- Conversational and language practice

- Educational content and personalized learning
- Translation services

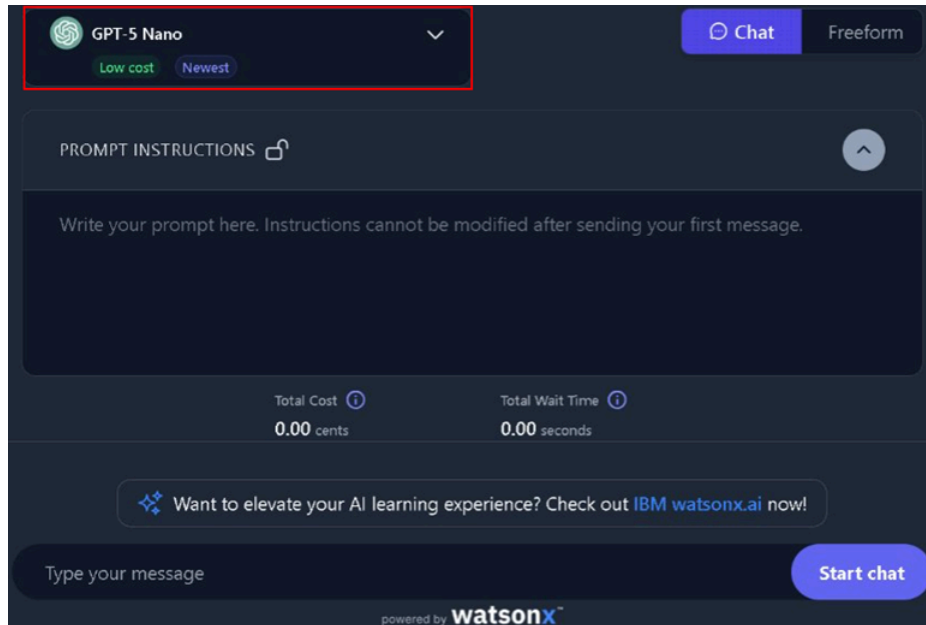
Let's begin experimenting with the foundation model.

Exercise 1: Learn a language and expand your vocabulary

In this exercise, you will learn vocabulary generation and language learning using the foundation model **GPT-5 Nano**.

Step 1: Set your AI environment and select the foundation model

1. As a first step, you can name the chat per your context.
2. Select the **GPT-5 Nano** model from the dropdown in the upper section of the right pane for this exercise.

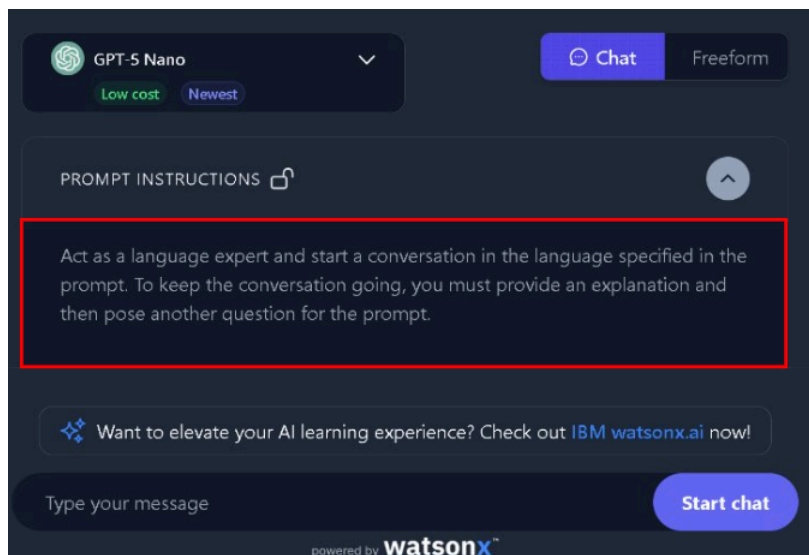


Step 2: Enhance your vocabulary in a familiar language

1. Consider a context to learn a language and enhance your vocabulary.
2. Provide the instructions in the **Prompt instructions** field.
For example,

Act as a language expert and start a conversation in the language specified in the prompt. To keep the conversation going, you must provide an explanation and then po

You can copy the above example using the copy icon on the right side of the box and paste it in the **Prompt instructions** field.



Note: You can also include instructions about the topic you want to learn.

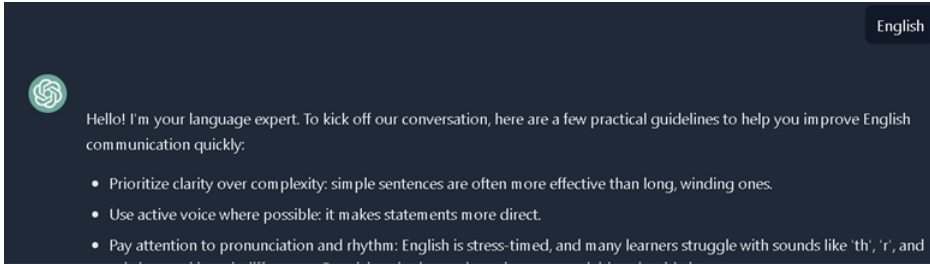
3. Write a prompt in **Type your message** field. Here, you can enter a language for which you want to enhance your vocabulary. For example,

English

Copy the above example using the copy icon on the right side of the box.

Note: You can specify any language of your choice.

4. Click **Start Chat** to get the chatbot response as a result in the middle part of the right pane. You will notice the generated response is similar to the screenshot below.

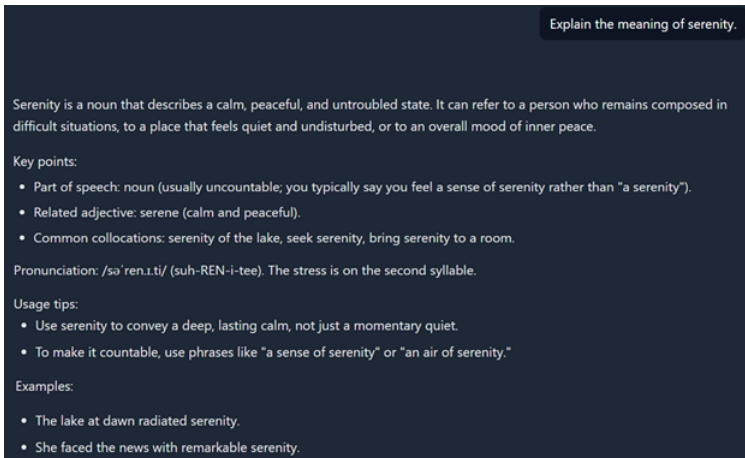


5. You can continue to converse with the system by responding to the query posed by the model and providing the context of the topic you want to learn about in the **Type your message** field.

For example,

Explain the meaning of serenity.

You will get a response similar to the screenshot below.



6. Next, you can continue to interact with the system and pose questions like

- Give me the synonyms of serenity.
- Give me the antonyms of serenity.
- When should one refrain from using the word serenity?
- Give me a fill in the blank or MCQ type question to test my knowledge of serenity word usage.

You will get a response followed by a question that you can utilize to learn and practice the target language. The generated responses to some of the questions are shown below.

Give me the synonyms of serenity.



Here are common synonyms for serenity. They share the idea of calm, peace, or composure, but each carries slightly different nuance.

Common synonyms

- serenity
- calmness
- peacefulness
- tranquility
- quiet
- stillness

Give me a fill in the blank or MCQ type question to test my knowledge of serenity word usage.



Hello! Here are a few MCQ and fill-in-the-blank questions to test your knowledge of serenity word usage.

Question 1: MCQ

- Which option best describes serenity?
- A) Absence of challenges
- B) Intense happiness at all times
- C) Inner calm, emotional balance, and a clear response to circumstances
- D) Pure external calm

Answer: C Rationale: Serenity refers to an inner state of calm and balanced response to life's challenges.

Question 2: Fill in the blank

Serenity involves recognizing what is within your control and choosing to respond rather than ____.

Step 3: Review results and responses

1. Continue interacting with the chatbot to generate more and more responses. The responses followed by a question can be utilized to learn and practice the target language.
2. Use the **Regenerate response** option to get another version of the responses to your latest question.

Note: Observe how the foundation model **Gpt-5 Nano** helps enhance your language and vocabulary. This capability can be used to develop apps focused on language learning, and other language or content generation services.

Exercise 2: Learn a foreign language using foundation model

Let's now experiment with the foundation model and explore the capability of learning a foreign language, which you can further utilize to develop an AI app or embed and add to your existing app.

In this exercise, you will learn a foreign language using the foundation model **GPT-5 Nano**.

Step 1: Set your AI environment and select the foundation model

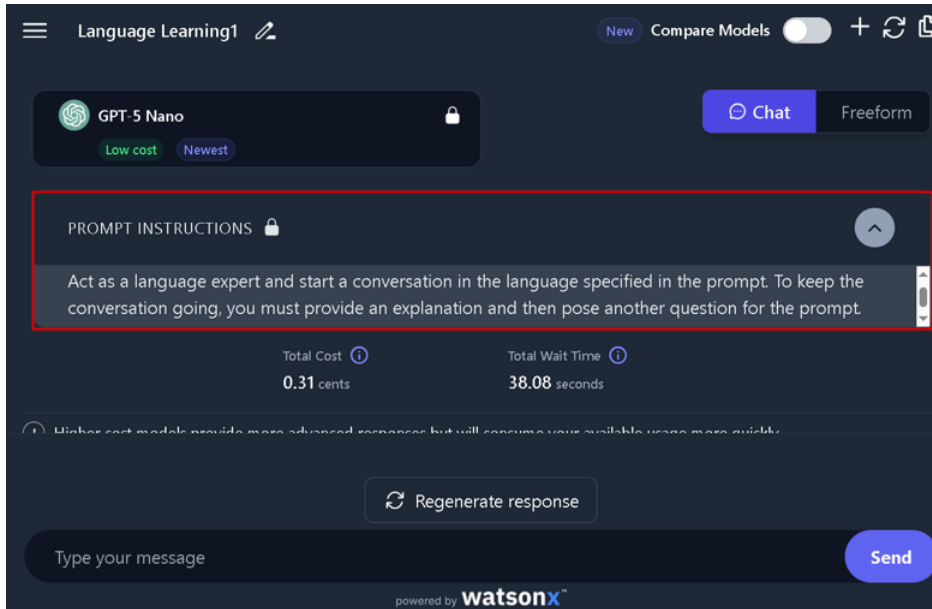
1. Name the chat and select the foundation model. You can create a new chat or continue with the same chat from exercise 1.
 - a. You can create a new chat and select **GPT-5 Nano** as the foundation model.
 - b. Alternatively, continue with the same chat under exercise 1 to use the same prompt instructions.
- You can choose **Reset chat** if you want to remove the previous conversation under the same chat window.
- You can also click **Duplicate chat** to create a copy of the existing chat and modify the prompt instructions.

Step 2: Prompt the model to learn a foreign language

1. Think of a foreign language you want to learn.
2. Provide the instructions in the **Prompt instructions** field.
For example,

Act as a language expert and start a conversation in the language specified in the prompt. To keep the conversation going, you must provide an explanation and then po

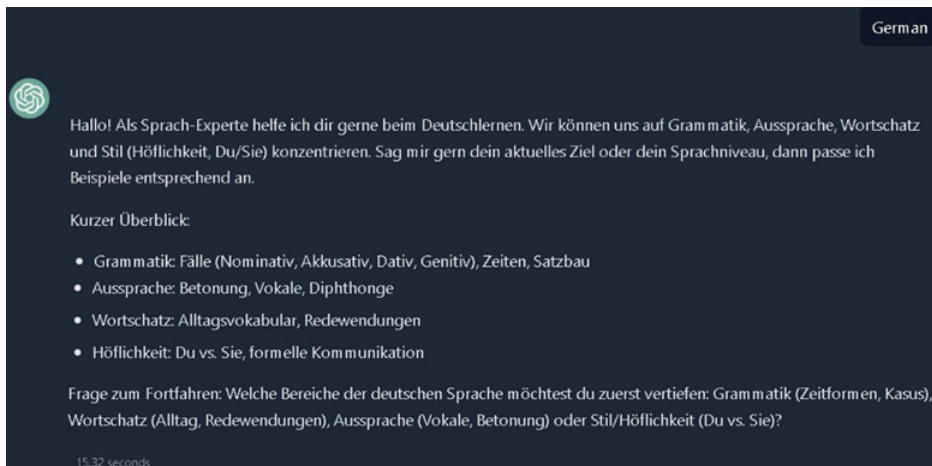
You can copy the above example using the copy icon on the right side of the box and paste it in the **Prompt instructions** field.



3. Write the language you want to learn in the **Type your message** field.
For example,

German

4. Click **Start Chat** and observe the generated response, similar to the screenshot below.



5. Continue to converse with the system by responding to the query posed by the model and providing the context of the topic you want to learn about in the **Type your message** field.
For example,

What is serenity in German?

You will get a response similar to the screenshot below.



Serenity bedeutet auf Deutsch am ehesten Gelassenheit, innere Ruhe oder Seelenfrieden. Je nach Kontext passen auch andere Ausdrücke wie friedlich oder heiter. Hier eine kurze Orientierung:

- Häufige Übersetzungen:
 - Gelassenheit: Ruhe und Gleichmut in schwierigen Situationen.
 - Seelenfrieden: innerer Frieden, oft nach innerem Aufruhr.
 - Innere Ruhe: allgemeine, anhaltende Ruhe im Inneren.
- Adjektivische Entsprechungen (für das englische Adjektiv serene):
 - gelassen, ruhig, heiter
- Beispielsätze:
 - Die Gelassenheit in ihrem Ton war deutlich zu hören.
 - Nach der schwierigen Zeit suchte er Seelenfrieden.
 - Er bewahrt in allen Situationen innere Ruhe.

6. Next, you can continue to interact with the system and ask questions like

- Tell me the synonyms and their usage of the word Gelassenheit.
- Give me the antonyms of serenity.
- Teach me German idioms or expressions related to emotions or feelings.
- Give me a fill-in-the-blank or MCQ-type question to test my knowledge of the word usage.

The generated responses to some of the questions are shown below.

Tell me the synonyms and their usage of the word Gelassenheit.



Gern! Hier ist eine kompakte, gut strukturierte Übersicht zu Synonymen von Gelassenheit und deren typischer Verwendung im Deutschen.

Wichtige Synonyme und ihre Nuancen

- Ruhe: Allgemeine Ruhe, sowohl äußere Stille als auch innere Ruhe; oft als neutrale Grundhaltung verwendet. Beispiel: "Er behielt in der hitzigen Debatte seine Ruhe."
- Gleichmut: Gleichmäßige innere Haltung, Stress oder Krisen wenig bis gar nichts aus dem Gleichgewicht bringend; oft philosophisch oder stilvoll. Beispiel: "Sie begegnete der Kritik mit Gleichmut."
- Ausgeglichenheit: Emotionales Gleichgewicht und Beständigkeit über verschiedene Situationen hinweg; betont stabile Persönlichkeit. Beispiel: "Die Ausgeglichenheit ihres Wesens zeigte sich im Alltag."
- Besonnenheit: Überlegtes, vorsichtiges Handeln bzw. Entscheiden; betont kognitive Ruhe und Abwägung. Beispiel: "Er traf die Entscheidung mit Besonnenheit."
- Souveränität: Selbstbewusstes, kontrolliertes Auftreten; verbindet Gelassenheit mit Kompetenz und Führungsqualität. Beispiel: "Sie trat mit Souveränität auf, trotz der Prüfung."
- Innere Ruhe / Seelenfrieden: Tiefe, oft andauernde Ruhe im Inneren; etwas poetischer oder spiritueller konnotiert. Beispiele: "Nach der Meditation verspürte er innere Ruhe." / "Nach der Klärung fand er Seelenfrieden."

Teach me German idioms or expressions related to emotions or feelings.



Hallo! Hier ist eine kompakte Sammlung gängiger deutscher Redewendungen und Idiome zu Gefühlen und Emotionen. Jede Angabe enthält die Bedeutung und ein Beispiel, damit du sie im Alltag leicht anwenden kannst. Hinweis: Viele Ausdrücke sind eher umgangssprachlich oder stilistisch; in formellen Texten solltest du sie sparsam verwenden.

Lass uns danach gemeinsam festlegen, welche Bereiche du zuerst vertiefen möchtest. Möchtest du dich auf Liebe/Romantik, Ärger/Frustration, Neugier/Begeisterung oder allgemeine Gefühlsausdrücke konzentrieren? Falls du magst, poste einfach ein Beispiel, das du übersetzen oder üben willst.

Positive Gefühle

- Schmetterlinge im Bauch haben — bedeutet: nervös-aufgeregt vor Liebe oder großer Vorfreude sein.
 - Beispiel: Seit dem ersten Date habe ich Schmetterlinge im Bauch.
- Auf Wolke sieben schweben — extrem glücklich sein, in einer gefühlsbetonten Euphorie.
 - Beispiel: Seit dem Treffen mit ihr schwebt er auf Wolke sieben.
- Jemandem das Herz erwärmen — eine Geste, die warme Gefühle auslöst.
 - Beispiel: Ihre Nachricht hat mir das Herz erwärmt.

Step 3: Review results and responses

1. Continue interacting with the chatbot to generate more and more responses. The responses followed by a question can be utilized to learn and practice the target language.
2. Use the **Regenerate response** option to get another version of the responses to your latest question.

Note: Observe how the foundation model **GPT-5 Nano** helps with language learning and practice. This capability can be used to develop apps focused on language learning, translation, and other language or content generation services.

Exercise 3: Translate text using a foundational model

Let's explore the translation capabilities of foundation models that can help you develop an app assisting language translation.

In this exercise, you will use the foundation model **GPT-5 Nano** to translate the text.

Step 1: Set your AI environment and select the foundation model

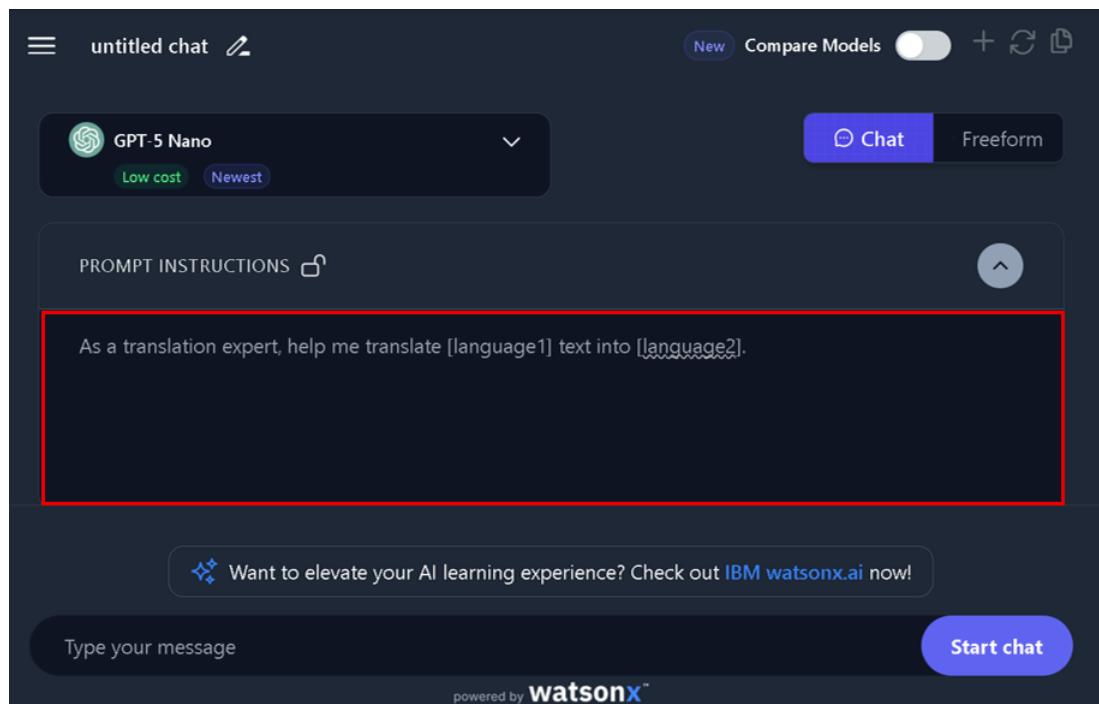
1. Create a new chat.
2. Name the chat per the context.
3. Select the foundation model **GPT-5 Nano**.

Step 2: Write text to translate from one language to another

1. Consider the text you need to translate.
2. Provide the instructions in the **Prompt instructions** field.
For example,

As a translation expert, help me translate [language1] text into [language2].

You can copy the above example using the copy icon on the right side of the box and paste it in the **Prompt instructions** field.



3. Provide language1 and language2 as prompts in the **Type your message** field.
For example,
English, German

4. Click **Start chat** to get a response like “Sure, I can help you with that. Please provide the text you would like to translate from English to German.” as shown below.



I can translate from English to German. Please paste the English text you want translated. To tailor the translation, please specify:

- Tone: formal (Sie) or informal (du)
- Region variant: de-DE (Germany), de-AT (Austria), de-CH (Switzerland) or general German
- Terminology: brand names, technical terms to preserve
- Audience: general public, professionals, children, etc.

If you prefer a neutral German suitable for most contexts, I can default to a formal tone and standard vocabulary.

Tips:

- Keep source text free of idioms that don't translate well; I can adapt them.
- Include any formatting (headings, bullet points) to preserve structure.

Paste the English text here and indicate preferences, and I will return the German translation.

5. Now, write the text you want to translate from one language to the target language.
For example,

Please convert the poem "Nothing Gold Can Stay" given below:
Nature's first green is gold,
Her hardest hue to hold.
Her early leaf's a flower;
But only so an hour.
Then leaf subsides to leaf.
So Eden sank to grief,
So dawn goes down to day.
Nothing gold can stay.

6. Click **Start chat** and review the generated response shown below.

Please convert the poem "Nothing Gold Can Stay" given below. Nature's first green is gold, Her hardest hue to hold. Her early leaf's a flower; But only so an hour. Then leaf subsides to leaf. So Eden sank to grief, So dawn goes down to day. Nothing gold can stay.



Das erste Grün der Natur ist Gold, der schwerste Farbton, den sie festhalten kann. Ihr frühes Blatt ist eine Blume; Doch nur eine Stunde lang. Dann schwindet das Blatt zum nächsten Blatt. So sank Eden in Trauer, So geht der Morgen dem Tag entgegen. Nichts Goldenes kann bleiben.

Hinweis zur Übersetzung

Diese Übersetzung bewahrt Bildsprache und Stimmung des Originals. Die Zeilenlängen wurden so angepasst, dass Sinn, Rhythmus und Klang erhalten bleiben.

Anmerkungen

- Kernbilder: Grün, Gold, Blätter, Eden, Morgendämmerung.
- Stil: poetisch; Reim und Metrik wurden zugunsten Verständlichkeit leicht angepasst.
- Varianten: 'Nichts Goldenes kann bleiben' ist eine gebräuchliche Formulierung; 'Nichts Goldes kann bleiben' ist ebenfalls korrekt; Wahl je nach bevorzugtem Klang.

37.09 seconds

Note: You can continue translating more and more text as required.

Step 3: Translate into a different target language

1. To translate text into another target language, reset the chat and enter the base language1 and the target language2 in the **Type your message** field.
For example,

English, French

2. Click **Start chat** and review the generated response, as shown below.

English, French



Voici une traduction française neutre et poétique du poème Nothing Gold Can Stay:

Le premier vert de la nature est l'or, Sa teinte la plus difficile à maintenir. Son premier feuillage est une fleur; Mais seulement pendant une heure. Puis la feuille cède sa place à une autre feuille. Ainsi Éden sombra dans le chagrin, Ainsi l'aube cède le pas au jour. Rien de doré ne peut durer.

Notes de traduction

- Ton: neutre et poétique, adapté au public général.
- Version française: l'Éden peut être écrit Éden ou Eden; j'ai utilisé Éden pour la translittération poétique.
- Préférence de registre: standard français, sans marque régionale spécifique.

25.37 seconds

3. Now, write the text you want to translate from one language to another.
For example,

Please translate the poem "The road not taken" as given below:
Two roads diverged in a yellow wood,
And sorry I could not travel both
And be one traveler, long I stood
And looked down one as far as I could
To where it bent in the undergrowth;
Then took the other, as just as fair,
And having perhaps the better claim,
Because it was grassy and wanted wear;
Though as for that the passing there
Had worn them really about the same
And both that morning equally lay
In leaves no step had trodden black.
Oh, I kept the first for another day!
Yet knowing how way leads on to way,
I doubted if I should ever come back.
I shall be telling this with a sigh
Somewhere ages and ages hence:
Two roads diverged in a wood, and I—
I took the one less traveled by,
And that has made all the difference.

4. Click **Start chat** and review the generated response shown below.



Deux chemins se séparèrent dans un bois jaune, Et, hélas, je ne pouvais pas tous les emprunter, Et étant seul voyageur, longtemps je me tins là, Et regardai l'une aussi loin que je pus, Jusqu'à l'endroit où elle se courbait dans le sous-bois; Puis pris l'autre, tout aussi juste, Et ayant peut-être le meilleur droit, Parce qu'elle était herbeuse et peu usée; Bien que, en ce qui concerne cela, le passage là-bas Les avait vraiment usés à peu près de la même manière; Et les deux, ce matin-là, gisaient également Dans des feuilles qu'aucun pas n'avait noircies. Oh, j'ai gardé le premier pour un autre jour ! Pourtant, sachant que tout chemin mène à d'autres chemins, Je doutai de pouvoir jamais revenir. Je raconterai cela avec un soupir Quelque part, dans des siècles et des siècles à venir: Deux chemins se séparèrent dans un bois, et moi — J'ai pris celui qui était le moins fréquenté, Et cela a fait toute la différence.

Notes de traduction

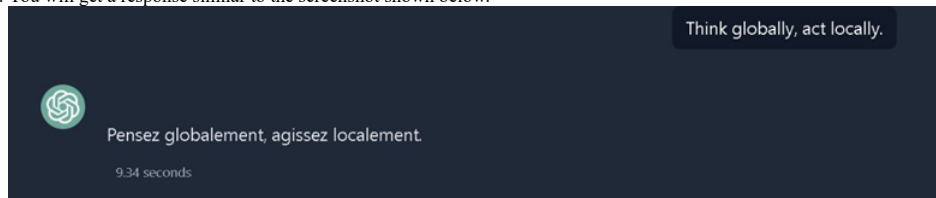
Traduction en français neutre et poétique qui respecte le sens, les images et le rythme de l'original. L'alignement ligne par ligne cherche à préserver la structure du poème.

40.91 seconds

5. Now try with another text, such as:

Think globally, act locally.

6. You will get a response similar to the screenshot shown below.



Step 4: Review results and responses

1. Continue interacting with the chatbot to translate more and more text from one language to another.
2. Use the **Regenerate response** option to get another version of the responses to your latest question.

Note: Observe how the foundation model **GPT-5 Nano** helps with translation and language learning. This capability can be used to develop apps focused on language translation services.

Summary:

Congratulations! You just completed the hands-on lab to develop AI applications using foundation models.

In this lab, you used foundation models of generative AI for vocabulary enhancement, language learning, and language translation. You used various prompts to experiment with various foundation models using the generative AI classroom lab. You also generated content for various natural language processing (NLP) tasks, including text and content generation, text translation, and code generation.

To leverage GPT-5 Nano for these purposes, you must use the OpenAI API and integrate it into your language learning or AI app development project. The API allows you to interact with the model and access its language generation capabilities programmatically, making it easier to create tailored language learning experiences or develop AI-powered language-related applications.

© IBM Corporation. All rights reserved.